Three-Point Finite-Difference Differentiation

Plus Composite Trapezoidal and Simpson Integration

Carson Quesada, Ishmael Taleo, Juan Ortiz,
Michael Lanfranchi, Rushi Sharma, Timothy Jung

- Approximate first derivatives from sampled data using three-point stencils.
- Validate central, forward, and backward formulas against analytic derivatives.
- Validate composite trapezoidal and Simpson rules on an integral with exact value π.
- Apply the differentiation workflow to estimate g from free-fall position-time measurements.

- Central:

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}$$

- Forward:

$$f'(x) \approx \frac{-3f(x) + 4f(x+h) - f(x+2h)}{2h}$$

- Backward:

$$f'(x) \approx \frac{f(x-2h) - 4f(x-h) + 3f(x)}{2h}$$

- Test functions: $\sin(x)$ at $\pi/4$, e^x at 0.3, and $x^3 - 2x^2 + x - 5$ at 1.2.
- Step size: $h = 10^{-5}$.
- All 9 method-function combinations passed tolerance with errors below 1.5×10^{-10}.

Function and point	Method	Abs. error
$\sin(\pi/4)$	central	1.40×10^{-11}
$e^{0.3}$	forward	3.38×10^{-11}
$p'(1.2)$	backward	1.40×10^{-10}

- Trapezoidal:

$$\int_a^b f(x)\,dx \approx h\left[\frac{f(x_0)+f(x_n)}{2}+\sum_{i=1}^{n-1} f(x_i)\right]$$

- Simpson (even n):

$$\int_a^b f(x)\,dx \approx \frac{h}{3}\left[f(x_0)+f(x_n)+4\sum_{i=1,\,\text{odd}}^{n-1} f(x_i)+2\sum_{i=2,\,\text{even}}^{n-2} f(x_i)\right]$$

- Benchmark integral:

$$\int_0^1 \frac{4}{1+x^2}\,dx = \pi$$

- Subinterval sweep: $n = 4, 8, 16, 32, 64, 128, 256$ for both methods.
- Trapezoidal observed order: 1.99999 (expected 2).
- Simpson observed order: 6.65525 (expected 4; inflated by near-zero terminal errors).
- Finest-mesh errors: trapezoidal 2.543×10^{-6}, Simpson 0 (within machine precision).

Method	Expected order	Observed order	Final tolerance pass
Trapezoidal	2	1.99999	Yes
Simpson	4	6.65525	Yes

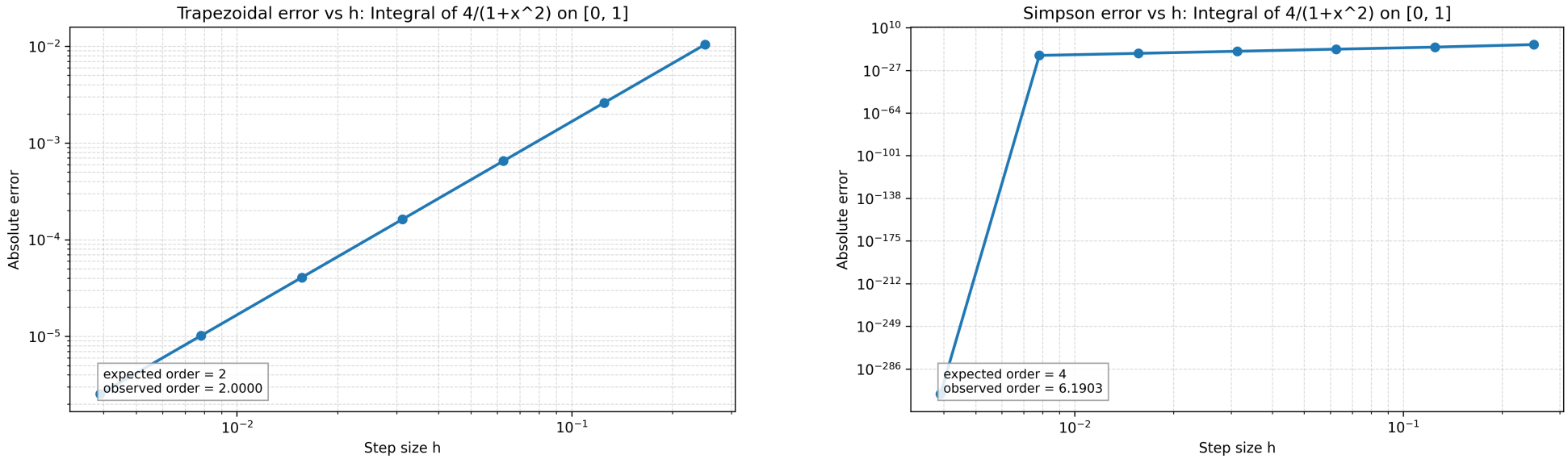

Figure 1: Log-log absolute error trends for composite trapezoidal and Simpson rules on the π benchmark.

- Fit a quadratic interpolant to 7 measured position-time samples.
- Differentiate numerically twice using the central stencil.
- Evaluate at $t_0 = 0.174306\,\mathrm{s}$ to recover acceleration.

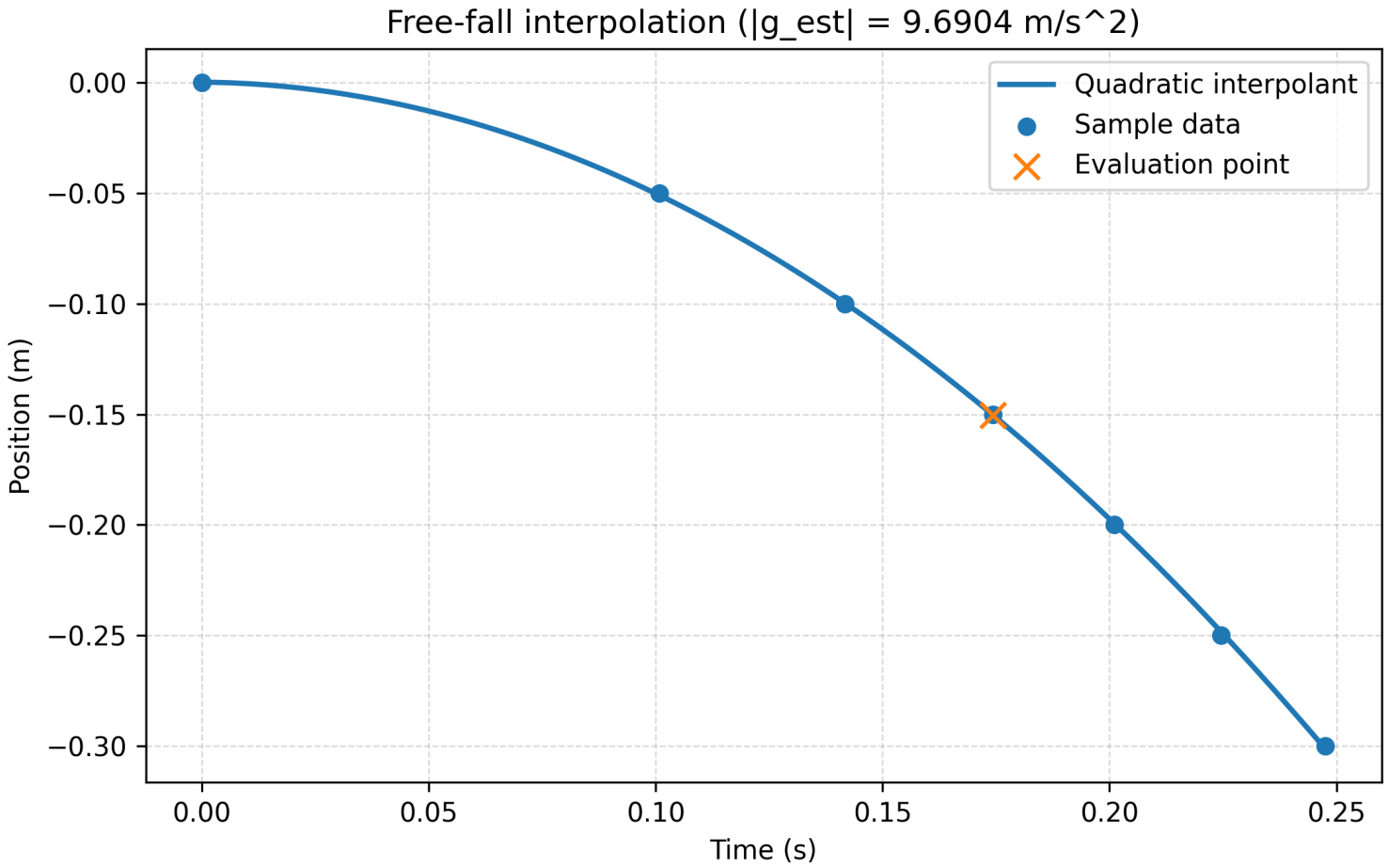

Figure 2: Quadratic interpolant fitted to measured free-fall data.

- Signed estimate: $\hat{a} = -9.6904\,\mathrm{m/s^2}$ (downward direction).
- Magnitude: $|\hat{g}| = 9.6904\,\mathrm{m/s^2}$.
- Reference: $9.81\,\mathrm{m/s^2}$.
- Absolute magnitude error: $0.1196\,\mathrm{m/s^2}$.
- Tolerance check: pass ($0.1196 < 0.15$).

- **differentiation-3point**: Builds and validates 3-point finite-difference workflows, exports CSV/JSON/Markdown outputs, generates plots, and estimates free-fall gravity.
- **integration-trapezoidal**: Implements composite trapezoidal integration with benchmark tests, convergence checks, and report-ready exports.
- **integration-simpson**: Implements composite Simpson integration with even-subinterval validation, fourth-order convergence analysis, and exported artifacts.

- **latex-article**: Generates IS&T-style technical articles from a target subject using the course template and required structure.
- **latex-slides**: Builds concise unit slide decks by summarizing the corresponding article and reusing unit result figures.
- These skills standardize file locations, output artifacts, and formatting so results remain reproducible across assignments.

- Agents are specialized assistants configured in `.github/agents/`.
- They apply focused instructions, tools, and constraints for different task types.
- In this project, one agent is centered on LaTeX formatting and one on Python numerical workflows.

- **ECE5110 LaTeX Formatter Agent**: Maintains consistent article/slide structure, styling, and build integrity; follows course templates and formatting constraints.
- **ECE5110 Python Numerical Agent**: Implements and validates numerical methods in `lib/` and `unit03/test/`, produces reproducible result artifacts and plots.
- Combined workflow: skills define reusable task recipes, while agents enforce role-specific quality and project conventions.

- Three-point finite-difference methods achieved strong analytic accuracy.
- Central differencing provided the best overall error performance.
- Trapezoidal and Simpson integration both passed convergence and final-accuracy checks.
- Simpson reached near machine precision on the smooth benchmark integral.
- Interpolation plus numerical differentiation recovered g within 1.2% of the standard value.

Questions?